

Practical: 2

Implementation and Time analysis of linear and binary search algorithm

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Aim: To Implement Linear search and binary search and Time analysis their algorithms.

Algorithm:

→ Algorithm:-

I) Linear Search :-
 Input : An Array
 Output : Target Element
 Step: 1 :- Start
 Step: 2 :- Read the array and the target element t.
 Step: 3 :- Set i = 0
 Step: 4 :- Compare array[i] with t
 Step: 5 :- If array[i] == t, return the index.
 Step: 6 :- Otherwise increment i and repeat step-4.
 Until the array ends
 Step: 7 :- If element Not found then
 return -1
 Step: 8 :- Stop

II) Binary search:-
 Input : Sorted Array
 Output : Target Element
 Step: 1 :- Start
 Step: 2 :- Read a sorted array and target element t
 Step: 3 :- Set l = 0 and h = n-1
 Step: 4 :- Calculate the middle position:

$$m = (l+h) / 2$$

 Step: 5 :- If arr[m] == t then
 return m
 Step: 6 :- if arr[m] < t, set l = m+1;
 else set h = m-1
 Step: 7 :- Repeat while (l <= h).
 Step: 8 :- if not found return -1
 Step: 9 :- Stop.

Program:

```
#include <iostream>
using namespace std;
```

```
// Linear Search
int linear(int array[], int s, int t) {
    for (int i = 0; i < s; i++) {
```

```
        if (array[i] == t) {
            return i;
        }
    }
    return -1;
}

// Binary Search
int binary(int array[], int s, int t) {
    int l = 0;
    int h = s - 1;

    while (l <= h) {
        int m = (l + h) / 2;

        if (array[m] == t) {
            return m;
        }
        else if (array[m] < t) {
            l = m + 1;
        }
        else {
            h = m - 1;
        }
    }

    return -1;
}

// Display Array
void display(int array[], int n) {
    cout << "Array is:\n";

    for (int i = 0; i < n; i++) {
        cout << array[i] << " ";
    }

    cout << endl;
}

int main() {

    int array[] = {2, 6, 7, 9, 11};
    int n = sizeof(array) / sizeof(array[0]);

    int t = 6;
```

```
display(array, n);

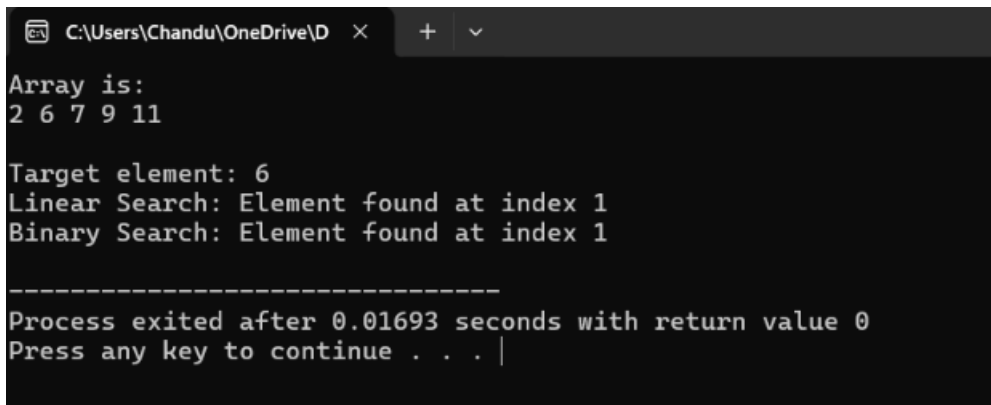
int f = linear(array, n, t);
int g = binary(array, n, t);

cout << "\nTarget element: " << t << endl;

if (f != -1) {
    cout << "Linear Search: Element found at index " << f << endl;
}
else {
    cout << "Linear Search: Element not found" << endl;
}

if (g != -1) {
    cout << "Binary Search: Element found at index " << g << endl;
}
else {
    cout << "Binary Search: Element not found" << endl;
}

return 0;
}
```

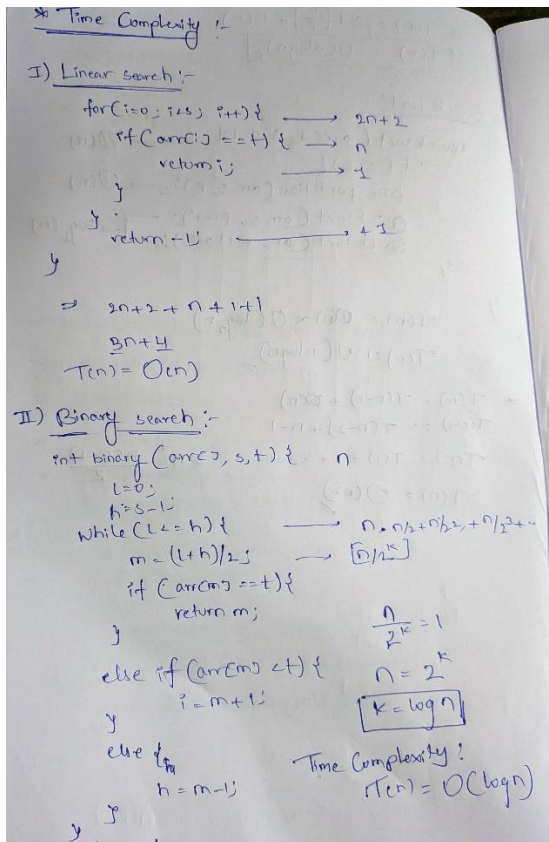
Output:

```
C:\Users\Chandu\OneDrive\D >
Array is:
2 6 7 9 11

Target element: 6
Linear Search: Element found at index 1
Binary Search: Element found at index 1

-----
Process exited after 0.01693 seconds with return value 0
Press any key to continue . . . |
```

Time Complexity:



Conclusion:

Hence

From this practical of linear search, Binary search the algorithm and programs have executed successfully. And Time complexity has categorised into three ways as Best, average and worst.